EX NO:2a	GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB
DATE:10/ 08/26	

Objectives
(i) To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.
(ii) To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule.

Objective 1: To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.

PROGRAM (MATLAB CODE)

```
clc;
 clear;
close all;
Am=1;
Ac=1;
fm=5;
fc=50;
beta=5;
kp=5;
fs=1000;
t=0:1/fs:1;
m = Am*cos(2*pi*fm*t);                % Message signal
c = Ac*cos(2*pi*fc*t);                % Carrier signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM
subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
```

Procedure:
1. Open MATLAB and create a new script file.
2. Clear the MATLAB workspace, command window, and close all existing figure windows using:
   clc; clear; close all;
3. Define the message signal parameters:
   - Message amplitude A_m =1

- Message frequency f_m=5Hz

4. Define the carrier signal parameters:
   - Carrier amplitude A_c=1
   - Carrier frequency f_c =50Hz
5. Set the modulation index β=5and sampling frequency fs=1000Hz.
6. Generate the time vector from 0 to 1 second with a sampling interval of 1/fs:

   t = 0:1/fs:1;
7. Generate the message signal using:

   m = Am*cos(2*pi*fm*t);

   This produces a low-frequency sinusoidal signal that acts as the modulating signal.
8. Generate the carrier signal using:

   c = Ac*cos(2*pi*fc*t);

   This produces a high-frequency sinusoidal carrier.
9. Generate the Frequency Modulated (FM) signal using:

   fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

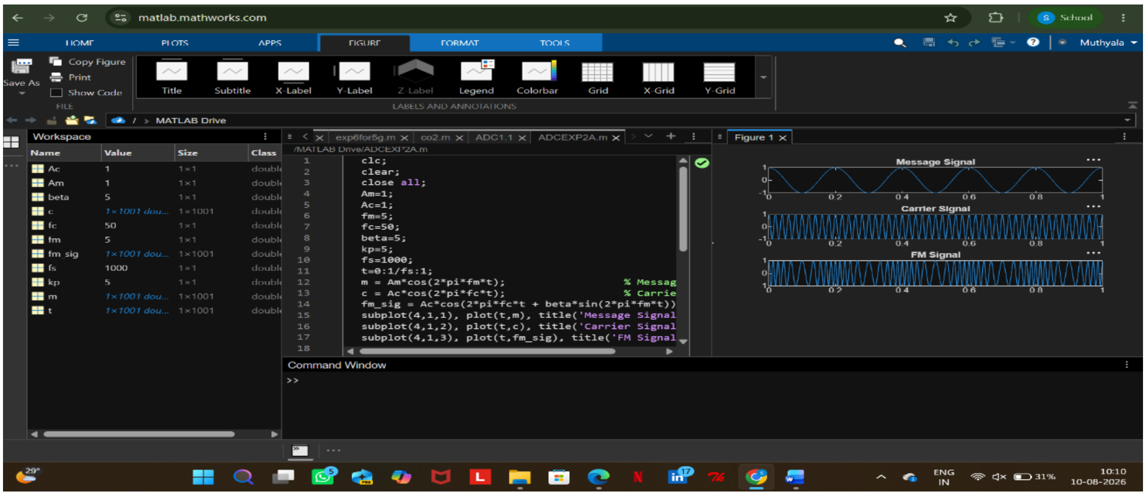
   Here, the instantaneous frequency of the carrier varies according to the amplitude of the message signal.
10. Plot the message signal using:

    subplot(4,1,1), plot(t,m), title('Message Signal')
11. Plot the carrier signal using:

    subplot(4,1,2), plot(t,c), title('Carrier Signal')
12. Plot the generated FM signal using:

    subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
13. Execute the program and observe the waveforms of the message signal, carrier signal, and FM signal.
14. Analyze the FM waveform and note that the frequency of the carrier varies according to the amplitude of the message signal while the amplitude of the FM signal remains constant.

OUTPUT (GRAPH)



Result

The Frequency Modulated (FM) signal was successfully generated using MATLAB, and the time-domain waveforms of the message signal, carrier signal, and FM signal were observed and analyzed.

QUESTION 1

A local FM radio station plans to transmit an audio signal of frequency 1 kHz. The station uses a carrier frequency of 100 kHz and a frequency sensitivity of 5 kHz/V. The audio signal amplitude is 2 V.

Tasks

1. Develop a MATLAB program to generate the message signal.
2. Generate the corresponding FM signal.
3. Plot the message and FM waveforms.
4. Calculate the maximum frequency deviation.
5. Determine the modulation index

QUESTION 2

A community radio station wants to estimate the bandwidth required for transmission. The station uses:

- Message frequency = 2 kHz
- Message amplitude = 3 V
- Carrier frequency = 75 kHz
- Frequency sensitivity = 3 kHz/V

Tasks

1. Generate the FM signal.
2. Calculate the maximum frequency deviation.
3. Determine the modulation index.
4. Calculate bandwidth using Carson's Rule.
5. Plot the waveform.

Objective 2: To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule

PROGRAM (MATLAB CODE)

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signals
clc;
clear;
close all;
%% Input Parameters
Am = 2;            % Message signal amplitude
fm = 500;          % Message frequency (Hz)
Ac = 1;            % Carrier amplitude
fc = 10000;        % Carrier frequency (Hz)
kf = 1000;         % Frequency sensitivity (Hz/V)
fs = 100000;       % Sampling frequency (Hz)
T = 0.02;          % Simulation duration (s)
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation Calculation
delta_f = kf * Am;
%% Modulation Index Calculation
beta = delta_f/fm;
%% FM Signal Generation
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Plot Message Signal
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
%% Plot FM Signal
subplot(2,1,2);
```

```
plot(t,fm_signal,'LineWidth',1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
%% Spectrum Analysis
N = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal))/N);
f = (-N/2:N/2-1)*(fs/N);
figure;
plot(f,FM_fft,'LineWidth',1.5);
title('FM Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([fc-10000 fc+10000]);
%% Display Results
fprintf('\nFM SIGNAL PARAMETERS\n');
fprintf('---------------------------\n');
fprintf('Carrier Frequency      = %.0f Hz\n',fc);
fprintf('Message Frequency      = %.0f Hz\n',fm);
fprintf('Message Amplitude      = %.2f V\n',Am);
fprintf('Frequency Sensitivity  = %.0f Hz/V\n',kf);
fprintf('\nCALCULATED VALUES\n');
fprintf('---------------------------\n');
fprintf('Frequency Deviation (Δf) = %.0f Hz\n',delta_f);
fprintf('Modulation Index (β)     = %.2f\n',beta);
%% Carson's Rule Bandwidth
BW = 2*(delta_f + fm);
fprintf('Bandwidth (Carson Rule) = %.0f Hz\n',BW);
```

Procedure

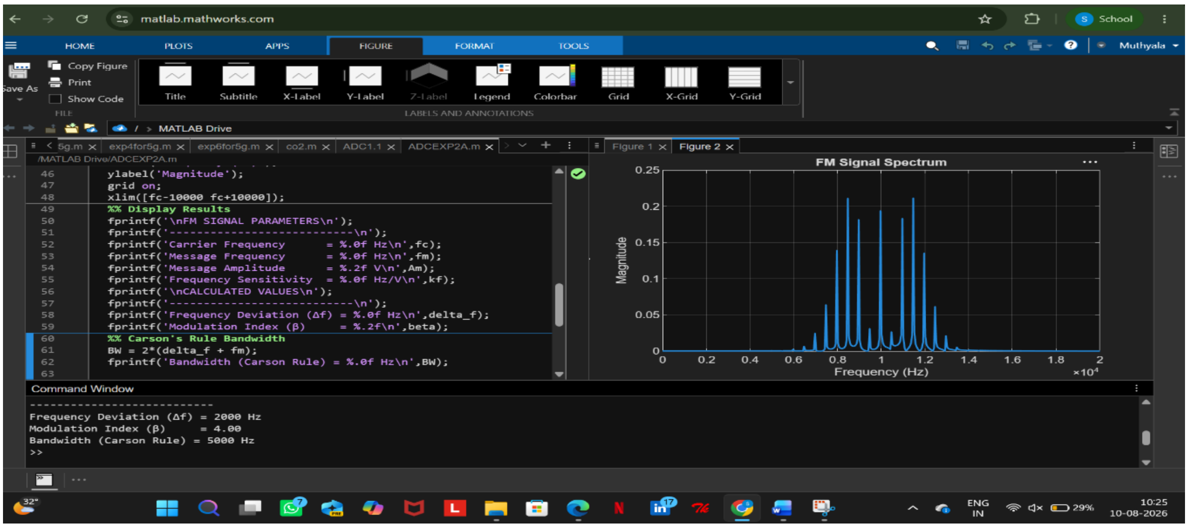
1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all existing figure windows using:

   clc;
   Clear;
   close all;

3. Define the message signal parameters:
   - Message amplitude Am=2V
   - Message frequency fm=500Hz
4. Define the carrier signal parameters:
   - Carrier amplitude Ac=1V
   - Carrier frequency fc=10000Hz

5. Set the frequency sensitivity constant kf=1000Hz/V, sampling frequency fs=100000Hz, and simulation duration T=0.02s.
6. Generate the time vector using:
   t = 0:1/fs:T;
7. Generate the message signal using:
   m = Am*cos(2*pi*fm*t);
8. Calculate the frequency deviation of the FM signal using:
   $\Delta f = kf.Am$
9. Calculate the modulation index using:
   $\beta = \Delta f/f\_m$
10. Generate the FM signal using:
   fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
11. Plot the message signal in the time domain and observe its waveform.
12. Plot the generated FM signal and observe that the carrier frequency varies according to the message signal while the amplitude remains constant.
13. Determine the length of the FM signal:
   N = length(fm_signal);
14. Compute the Fast Fourier Transform (FFT) of the FM signal:
   FM_fft = fft(fm_signal);
15. Shift the spectrum to center the zero-frequency component and normalize it:
   FM_fft = fftshift(abs(FM_fft)/N);
16. Generate the frequency axis corresponding to the FFT output:
   f = (-N/2:N/2-1)*(fs/N);
17. Plot the frequency spectrum of the FM signal and observe the carrier component and sidebands.
18. Restrict the frequency axis around the carrier frequency for better visualization:
   xlim([fc-10000 fc+10000]);
19. Calculate the bandwidth of the FM signal using Carson's Rule:
   $BW = 2(\Delta f + f\_m)$
20. Display the carrier frequency, message frequency, frequency deviation, modulation index, and bandwidth in the MATLAB command window.
21. Execute the program and verify that the calculated frequency deviation and modulation index match the values used in generating the FM signal

OUTPUT (GRAPH)



Example (Just for Understanding)

Given:

Message Amplitude A_m=2V

Message Frequency f_m=500Hz

Frequency Sensitivity k_f=1000Hz/V

Frequency deviation:

$\Delta f = k\_f\ A\_m$

$\Delta f = 1000 \times 2 = 2000$" Hz"

Modulation index:

$\beta = \Delta f / f\_m$

$\beta = 2000/500 = 4$

Bandwidth using Carson's Rule:

$BW = 2(\Delta f + f\_m) = 2(2000+500) = 5000$" Hz"

Result

i) The time-domain waveform was observed, and the frequency spectrum obtained using FFT has been plotted.

ii) Also, calculated the frequency deviation and bandwidth using Carson’s rule

EX NO: 2b	CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNALS USING MATLAB SIMULATION
DATE:10/ 08/2026	

Objectives:
To generate a Frequency Modulated (FM) signal in MATLAB using specified message and carrier signal parameters, and to determine the frequency deviation and modulation index of the FM signal while analyzing the relationship between frequency deviation, message frequency, and modulation index.

PROGRAM (MATLAB CODE)

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
 clc;
clear;
close all;
 %% Message Signal Parameters
Am = 2;          % Message amplitude (V)
fm = 500;        % Message frequency (Hz)
 %% Carrier Signal Parameters
Ac = 1;          % Carrier amplitude (V)
fc = 10000;      % Carrier frequency (Hz)
%% FM Parameters
kf = 250;        % Frequency sensitivity (Hz/V)
fs = 100000;     % Sampling frequency (Hz)
T = 0.01;        % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
 %% Modulation Index
beta = delta_f/fm;
 %% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
 %% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
```

```
 fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
fprintf('Modulation Index (Beta) = %.2f\n',beta);
 %% Verification
beta_verify = delta_f/fm;
 fprintf('Verified Modulation Index = %.2f\n',beta_verify);
 %% Verification Check
tolerance = 1e-6;
 if abs(beta - beta_verify) < tolerance
        fprintf('\nVerification Successful\n');
else
        fprintf('\nVerification Failed\n');
end
 %% Plotting
figure;
 subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
 subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
```

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all previously opened figure windows using:
   Clc;
   Clear;
   close all;
3. Define the message signal parameters:
   - Message amplitude A_m=2V
   - Message frequency f_m=500Hz
4. Define the carrier signal parameters:
   - Carrier amplitude A_c=1V
   - Carrier frequency f_c=10000Hz
5. Define the frequency sensitivity constant k_f, sampling frequency f_s, and simulation duration T
6. Generate the time vector using:

t = 0:1/fs:T;

7. Generate the message signal using:

   m = Am*cos(2*pi*fm*t);

   and observe the sinusoidal waveform.
8. Calculate the frequency deviation of the FM signal using:

   $\Delta f = kf. Am$

9. Calculate the modulation index using:

   $\beta = \Delta f / fm$

10. Generate the FM signal using:

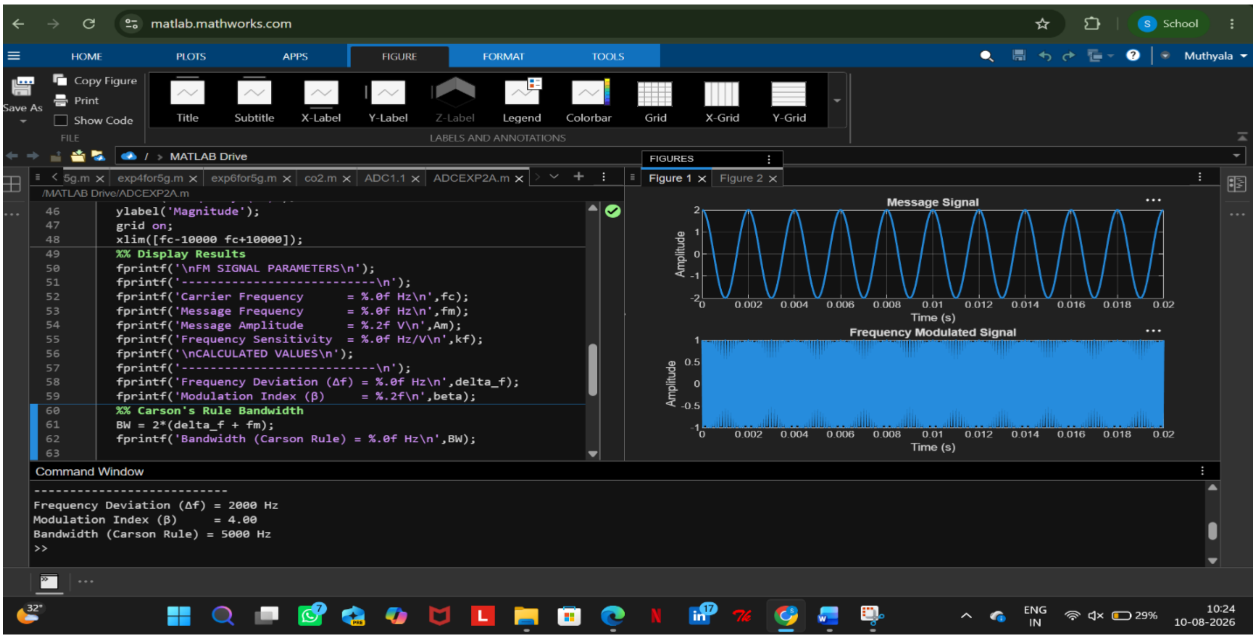
    fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

    where the carrier frequency varies according to the message signal.
11. Plot the message signal in the first subplot and label the axes appropriately.
12. Plot the FM signal in the second subplot and observe the variation in instantaneous frequency.
13. Display the carrier frequency, message frequency, message amplitude, and frequency sensitivity in the MATLAB command window using fprintf statements.
14. Display the calculated frequency deviation and modulation index values in the command window.
15. Verify the modulation index by recomputing it using:

    $\beta = \Delta f / fm$

    and compare the result with the previously calculated value.
16. Use a conditional statement (if-else) to check whether both modulation index values are equal within a small tolerance.
17. Display the verification result as:
    - "Verification Successful" if both values match.
    - "Verification Failed" otherwise.
18. Execute the MATLAB program and observe the message signal, FM signal, calculated frequency deviation, modulation index, and verification result.

OUTPUT GRAPH



Example (Just for Understanding)

For:

$A\_m=2V$

$f\_m=500Hz$

$k\_f=1000Hz/V$

Frequency deviation:

$\Delta f=k\_f\ A\_m$

$\Delta f=1000\times 2=2000" Hz"$

Modulation index:

$\beta=\Delta f/f\_m$

$\beta=2000/500=4$

Result

This experiment verifies that the modulation index is equal to the ratio of frequency deviation to message frequency, i.e., $\beta=\Delta f/f\_m$.